Mehmet Berk Gürçay

Senior Cloud & Backend Engineer

Munich, Germany · mbgurcay@gmail.com · linkedin.com/in/mehmet-berk-gurcay
github.com/mebegu · mebegu.github.io

SUMMARY

Backend engineer with 7+ years building serverless and event-driven systems on AWS, most recently on an IoT platform for more than 20 million connected tools and gateways. Earlier led the team behind a Client B CRM knowledge graph and built data pipelines and Power BI reporting for enterprise sales. Owns services through design, Terraform, and production debugging.

SKILL HIGHLIGHTS

- **AWS Serverless & Event-Driven**
  IoT Platform · Battery Quality · CRM KG
- **Solution Design & Architecture**
  IoT Platform · Battery Quality · Client D
- **IoT Platforms**
  IoT Platform (20M+ devices) · Client D
- **Data Engineering**
  Sales Analytics · CRM KG
- **Infrastructure as Code**
  IoT Platform · Client C · Battery Quality
- **Graph Data Modeling**
  CRM KG · Customer Feedback
- **Technical Leadership**
  CRM KG (8-person team, 3+ years)
- **Production Ops & Observability**
  IoT Platform · CRM KG · Sales Analytics
- **Agentic Software Development**
  IoT Platform · Battery Quality · Client C · SOC

Core tech: Python · TypeScript / Node.js · Docker & Kubernetes · GitLab CI/CD

EXPERIENCE

IoT Consultant · Concept Reply GmbH, Munich
2024-11 – now

- Designed and built OTA updates, device commands, and telemetry routing for Client A's AWS IoT platform (20M+ devices).
- Built the infrastructure, four-eyes approval, and audit trail for a Client A battery service as the sole Concept Reply engineer.
- Set up multi-account networking for the Client C AWS Landing Zone, developed an eBPF forensics pipeline for an open-source Kubernetes SOC, and built a GraphRAG chatbot at a Client B and Neo4j hackathon.

Tech Consultant · Motius GmbH, Munich
2019-05 – 2024-07

- Led an 8-person Scrum team for 3+ years on the Client B CRM knowledge graph: 19 analytical engines, Python ETL into Neo4j, and RAG with OpenAI.
- Built dbt models on Snowflake and Power BI reports for Client B sales leadership, and delivered projects for Client F, Client E, and a government office.

Co-Founder · Loopcake, Istanbul
2016-02 – 2017-12

- Built a Git-integrated learning platform; national entrepreneurship competition runner-up.

Project details with roles, team sizes, and full stacks follow on the next pages.

EDUCATION

Master of Science, Informatics · Technical University of Munich (TUM) — 2021

Bachelor of Science, Computer Engineering · Koç University, Istanbul — 2017

Internships: Client B R&D, Istanbul (Java plugin for JIRA) · İş Bank, Istanbul (network engineering)

CERTIFICATIONS

AWS Certified Solutions Architect – Associate — 2026

Neo4j Certified Professional — 2024

LANGUAGES

Turkish — Native

English — Advanced

German — Intermediate (written), pre-intermediate (spoken)

PROJECT DETAILS

IoT Consultant · Concept Reply GmbH, Munich

2024-11 – now

Client A – Battery Quality Service

Client: Client A, construction technology · **Role:** only Concept Reply engineer · **Team:** a Client A business department and Client A developers

A service that issues batch State-of-Health lock commands to batteries flagged by quality analysis, with four-eyes approval and a permanent audit trail.

- Built the Terraform infrastructure: Lambda, SQS with a dead-letter queue, DynamoDB, S3 with compliance-mode object lock, Secrets Manager, and IAM.
- Implemented four-eyes approval roles, lock issuance with prior and active lock checks, run counters, and audit records.
- Added structured logging, correlation IDs, CloudWatch alarms, an operations dashboard, and saved Logs Insights queries, and mapped legacy audit fields onto the new data model.
- Wrote the solution design for execution reporting and built the baseline report generator, working spec-driven with Kiro.

Stack: AWS Lambda · SQS · DynamoDB · S3 · CloudWatch · Terraform · Kiro · Kotlin · Quarkus

Client A – Next Generation IoT Platform

Client: Client A, construction technology · **Role:** core contributor · **Team:** 16 developers, including AWS Professional Services

A multi-account AWS-native platform to manage, update, and monitor more than 20 million connected power tools and gateways.

- Designed and built the over-the-air update pipeline with malware scanning, firmware signing, Step Functions orchestration, and fleet-wide job management.
- Developed the software update query service (compatibility evaluation, version comparison, OAuth scope-based visibility) and a signed device command interface on IoT device shadows with audit logging.
- Implemented an event-driven processor for device state changes and cross-account MQTT routing for parallel telemetry processing during the migration.
- Defined the Terraform for these services across 8 AWS accounts with several stages each and worked with AWS Professional Services on cross-account security.
- Debugged production issues with CloudWatch and X-Ray, and wrote requirements and designs as Kiro specs before implementation.

Stack: AWS Lambda · IoT Core · Step Functions · API Gateway · DynamoDB · SQS · Terraform · Python · Kiro

Client C – AWS Landing Zone Extension

Client: Client C, CAD/CAM software · **Role:** sole Concept Reply engineer, with a Reply product owner

Extending an existing AWS Landing Zone with more automation and consistent multi-account governance.

- Set up VPC and Transit Gateway structures for inter-account networking and supported the Route 53 delegation design.
- Automated provisioning with Terraform and GitLab CI/CD and documented account onboarding.

Stack: Control Tower · Organizations · Transit Gateway · Route 53 · Terraform · Claude Code · Codex

Adaptive Kubernetes SOC (open source)

Project: grant-funded open-source security project · **Role:** detection and forensics pipeline

A sovereign Kubernetes security operations center in which all security evidence stays in the operator's own infrastructure.

- Developed the eBPF-based pipeline (Pixie, Kubescape, Tetragon) that detects anomalies, triggers forensic traces, and builds STIX attack graphs.
- Designed Hive Sentinel, a Flask API that ingests telemetry through Pixie and Vector into ClickHouse, and built ETL pipelines for HTTP, DNS, and syscall logs.
- Maintained honeycluster, added STIX integration to Pixie, and contributed to a KubeCon Europe 2026 submission.

Stack: Kubernetes · Pixie · Tetragon · Kubescape · ClickHouse · Flask · GitHub Actions · Claude Code · Codex

GraphRAGical – Client B and Neo4j hackathon

Role: started the repository and built the chatbot · **Team:** about 7, including Client B and Neo4j engineers

A GraphRAG assistant that recommends Client B Xcelerator portfolio items based on a user's needs and industry.

- Built a two-stage LLM agent with an intent classifier on top of vector search and graph pattern matching in Neo4j.

Stack: Neo4j · neo4j-graphrag · Azure OpenAI · LangChain · Streamlit · Python

Client D – Device Management Platform

Client: Client D, industrial IoT for energy efficiency · **Role:** architecture review and device registry · **Team:** 4

Integrating IoT data from multi-protocol industrial devices into a cloud platform.

- Reviewed the existing architecture and proposed improvements.
- Migrated the device registry data layer to TypeORM, fixed device metrics upserts, introduced a shared Redis client, and automated ER diagram generation.

Stack: TypeScript · NestJS · TypeORM · PostgreSQL · Redis · Kafka · EMQX · MQTT · Sparkplug B · Azure DevOps

Tech Consultant · Motius GmbH, Munich

2019-05 – 2024-07

Client B – CRM Knowledge Graph (CRM KG)

Client: Client B · **Role:** backend lead, cloud engineer · **Team:** 8, Scrum

The knowledge graph behind a new CRM system for the sales department.

- Led an 8-person Scrum team for 3+ years, ran code reviews, trained juniors and interns, and set up Jest and SonarQube checks.
- Designed 19 analytical engines on AWS Glue, Lambda, SQS, and Kafka, with Python ETL from tabular data into a Neo4j graph model, and operated the pipelines in production.
- Synchronized data with a third-party vendor working on the same CRM system and implemented a RAG system with OpenAI and the Neo4j vector index.

Stack: Neo4j · Cypher · Python · AWS Glue · AWS Lambda · SQS · DynamoDB · Kafka · GraphQL · Jest

Client B – Sales Analytics

Client: Client B · **Role:** analytics engineer · **Audience:** sales teams, team leads, and senior management

Next-best-action recommendations for sales: trending products, potential customers, and high-profit sectors.

- Built dbt models and SQL transformations on Snowflake and the Power BI reports on recommendation accuracy and impact, published through Power BI Service, Teams, and SharePoint.
- Investigated failed loads and data quality issues in the existing ingestion pipelines and added smaller extensions.

Stack: Snowflake · dbt · SQL · Power BI · GitLab

Government – Project Management Platform

Client: Government office (2020) · **Role:** backend engineer · **Team:** 6, Scrum

A project management and communication tool with confidential reporting.

- Built a project management platform with task lifecycles, confidential PDF reports, and real-time notifications.

Stack: PostgreSQL · Express · React · Node.js · TypeScript · TypeORM · Swagger · Docker · GitLab CI/CD

Client B – Customer Feedback Service

Client: Client B · **Role:** backend engineer · **Team:** 6, Scrum

A catalog of customer feedback and follow-up actions for sales representatives.

- Designed a graph-based feedback system that lets representatives review historical feedback, with Python ETL into Neo4j and GraphQL APIs.

Stack: Neo4j · AWS Lambda · DynamoDB · S3 · TypeScript · GraphQL · Docker

Client E – In-Car Chassis Telemetry Games

Client: Client E · **Role:** Unity developer · **Team:** 3

Gamified infotainment screens driven by real-time chassis data.

- Built four Unity3D games and a menu hub driven by live chassis telemetry, shipped as an in-car demo.

Stack: Unity3D · C#

Client F – Car-Sharing Route Planner

Client: Client F · **Role:** sole developer, with a Motius product owner

Departure-time planning for a car-sharing platform.

- Built real-time car availability and departure-time planning from the client API and Google Maps traffic data.

Stack: Node.js · Express · TypeScript · MongoDB

SKILLS

AWS:	Lambda, IoT Core, Step Functions, API Gateway, ECS, Glue, DynamoDB, S3, RDS, SQS, SNS, KMS, CloudWatch, X-Ray, Bedrock, VPC, Transit Gateway, Route 53, Organizations, Control Tower, IAM Identity Center
Code:	Python, TypeScript, JavaScript, SQL, Cypher, C#, Kotlin, NestJS, Express, Flask, Quarkus, TypeORM, LangChain, Streamlit, Powertools for AWS Lambda, Pydantic
Data & messaging:	Snowflake, dbt, Neo4j, PostgreSQL, ClickHouse, MongoDB, Redis, Power BI, Kafka, MQTT, Sparkplug B, EMQX
Infrastructure:	Terraform, AWS CDK, CloudFormation, Docker, Kubernetes, Helm, GitLab CI/CD, GitHub Actions, Azure DevOps
Quality & AI:	pytest, Jest, Testcontainers, SonarQube, Kiro, Claude Code, GitHub Copilot, Codex

TEACHING & COMMUNITY

TUM teaching assistant, Software Project Organization and Management (2019) · KU-GameDev founder at Koç University, 13-person core team (2014 – 2017) · Neo4j for IoT and GenAI session, Concept Reply (2026)

INTERESTS

Writing a sci-fi fantasy novel · indie game development · chess and board games · space · football · cooking · travel